Mathematics for Cryptography: Basic Notation and Groups

October 29, 2026

Blockstream Research

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github.com/BlockstreamResearch/zk-book



Plan

1. Some words about the course
2. Notation
   - Sets
   - Logic
   - Randomness and Sequences
3. Basic Group Theory
   - Reasoning behind Groups
   - Group Definition and Examples
   - Subgroups
   - Cyclic Groups
   - Homomorphism and Isomorphism

Some words about the course

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Note

This course is helpful for everyone: even lecturers do not know all the material fully. Please, feel free to ask questions and provide feedback — we will adjust the material accordingly.

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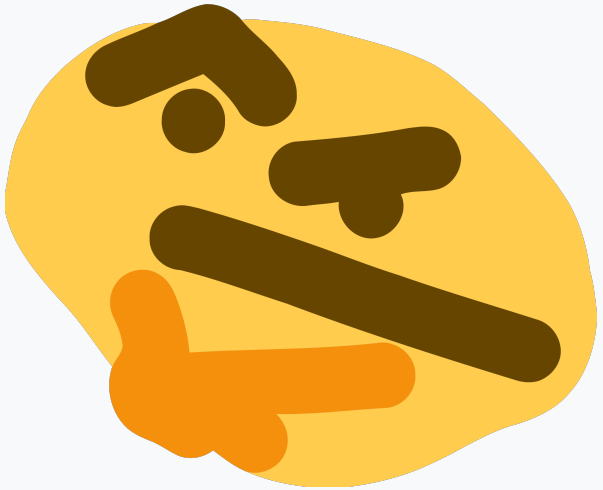
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Note

We are R&D experts in Cryptography, so we need to boost our skills in academic writing, lecturing, and understanding advanced topics.

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5. *Optionally*, we will conduct workshops on a separate day. We will discuss this later.

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4. Post-quantum zero-knowledge: lattice-based protocols (LaBRADOR) and hash-based ones (STARK).



Notation

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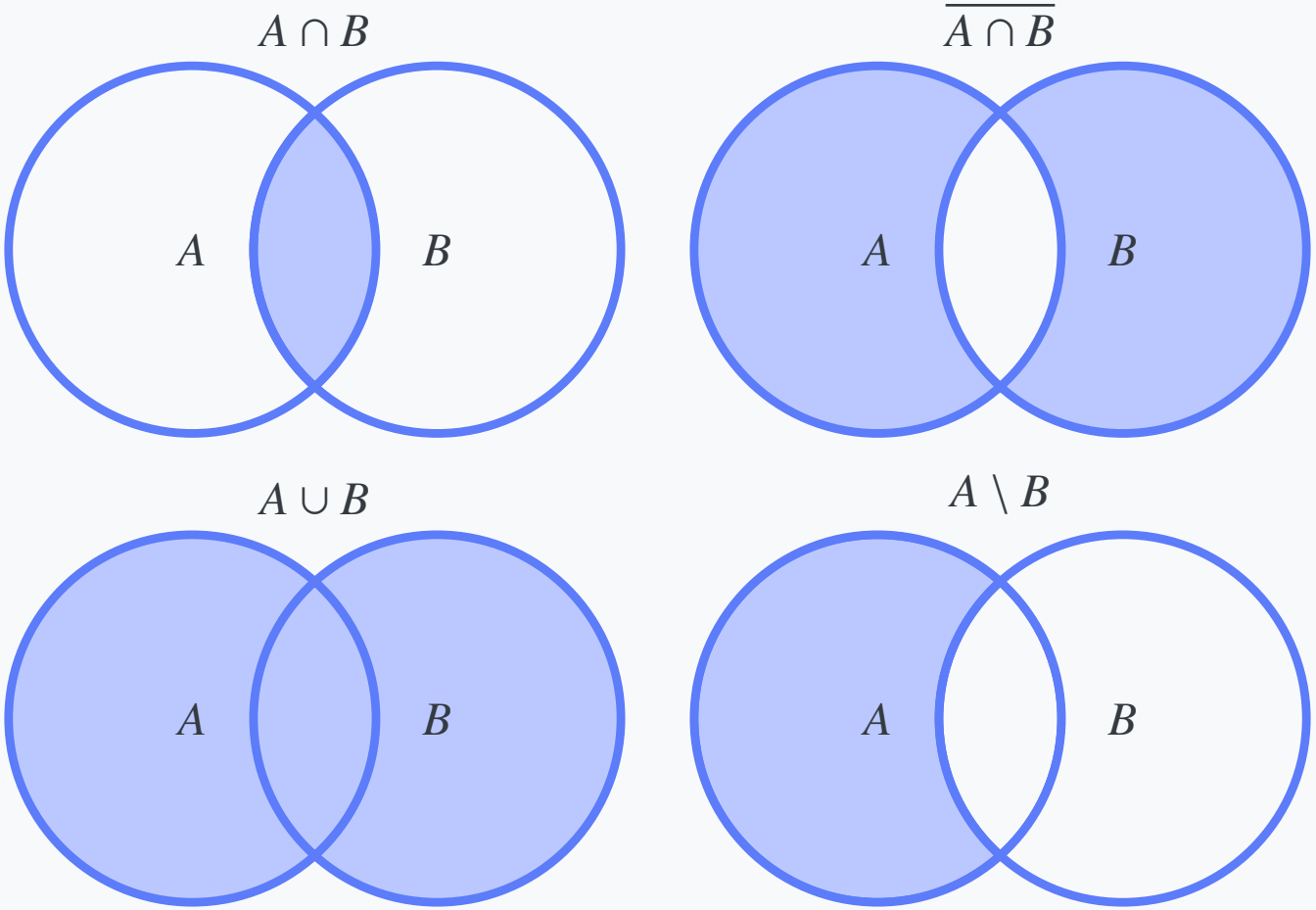
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- $\{1, 2, 2, 3\} = \{1, 2, 3\}$ — we do not count duplicates.
- $\{1, 2, 3\} = \{2, 1, 3\}$ — order does not matter.
- $\{\{1, 2\}, \{3, 4\}, \{\sqrt{5}\}\}$ — elements can be sets themselves.

Operations on sets



Operations on sets: Examples

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We write $\#A$ (or $|A|$) for the number of elements in A. For example, $\#\{1, 2, 5, 10\} = 4$.

Defining sets

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How to simplify the set $\{\sin \pi k : k \in \mathbb{Z}\}$?

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Example

$\mathbb{R}^2$ is a set of all possible points in the Cartesian plane.

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How to interpret the set $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$?

Basic Logic

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- $\exists$ means “there exists”, $\exists!$ means “there exists the only”.
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Is it true that $(\forall x \in \mathbb{Z})\,(\exists y \in \mathbb{N}) : \{y > x\}$?

Randomness and Sequences

Notation

To denote probability of event E, we use notation $\Pr[E]$. The probability of E given that F happened is $\Pr[E \mid F]$. For example,

$$\Pr[x \equiv 1 \pmod 3 \mid x \leftarrow \text{RollDice}()] = \frac{1}{3}.$$

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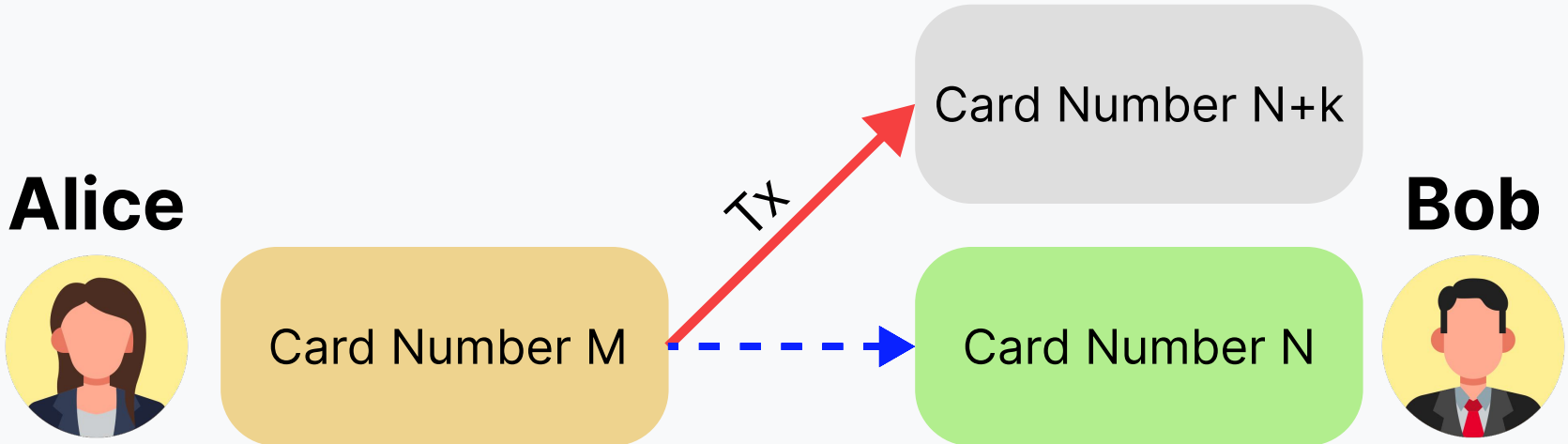
To denote an infinite sequence $x_1, x_2, \ldots$, we use $\{x_i\}_{i\in\mathbb{N}}$. We write $[n] := \{0, \ldots, n-1\}$, so a finite sequence $x_0, x_1, \ldots, x_{n-1}$ is $\{x_i\}_{i\in[n]}$.
To enumerate through a list of indices $\mathcal{I} \subseteq \mathbb{N}$, we use notation $\{x_i\}_{i\in\mathcal{I}}$.
Vectors are bold: $\mathbf{x} = (x_1, x_2, x_3) \in \mathbb{F}^3$.

Basic Group Theory

Why Groups?!

Well, first of all, we want to work with integers...

Imagine that Alice pays to Bob with a card number N, but instead of paying to a number N, the system pays to another card number $N + k, k \ll N$, which is only by 0.001% different. Bob would not be 99.999% happy...



Why Groups?!

But integers on their own are not enough. We need to define a structure that allows us to perform operations on them.

This is very similar to interfaces: we abstract from the implementation, just merely stating we have “some” operation.

Example

Consider set $\mathbb{G} := \{\text{Dmytro}, \text{Dan}, \text{Friendship}\}$. We can safely define an operation $\oplus$ as:

$$\text{Dmytro} \oplus \text{Dan} = \text{Friendship}$$
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Rhetorical question

What makes $(\mathbb{G}, \oplus)$ a group?

Group Definition

Definition

Group $(\mathbb{G}, \oplus)$, is a set with a binary operation $\oplus$ with following rules:

1. **Closure:** Binary operations always outputs an element from $\mathbb{G}$, that is $\forall a, b \in \mathbb{G} : a \oplus b \in \mathbb{G}$.

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Definition

A group is called **abelian** if it satisfies the additional rule called **commutativity**: $\forall a, b \in \mathbb{G} : a \oplus b = b \oplus a$.

Explanation for Developers: Trait

```rust
/// Trait that represents a group.
pub trait Group: Sized {
    /// Checks whether the two elements are equal.
    fn eq(&self, other: &Self) -> bool;
    /// Returns the identity element of the group.
    fn identity() -> Self;
    /// Adds two elements of the group.
    fn add(&self, a: &Self) -> Self;
    /// Returns the negative of the element.
    fn negate(&self) -> Self;
    /// Subtracts two elements of the group.
    fn sub(&self, a: &Self) -> Self {
        self.add(&a.negate())
    }
}
```

More on that: “Some Fun: Group Implementation in Rust” in the book (github.com/BlockstreamResearch/zk-book).

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Rule of thumb

We use additive notation when we imply that the group $\mathbb{G}$ is the set of points on the elliptic curve, while multiplicative is typically used in the rest of the cases.

Abelian Groups Examples and Non-Examples

Question #3

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Set V is a set of tuples (v_1, v_2, v_3) where each $v_i \in \mathbb{R} \setminus \{0\}$. Define the operation $\odot$ as

$$(v_1, v_2, v_3) \odot (u_1, u_2, u_3) = (v_1 u_1, v_2 u_2, v_3 u_3)$$

Is $(V, \odot)$ a group? If no, what is missing?

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Is $(V, \odot)$ a group? If no, what is missing?

Conclusion

Group is just a fancy name for a set with a binary operation that behaves nicely.

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Question #2*

Let $\mathrm{GL}(\mathbb{R}, 2)$ be a mutliplicative group of invertable matrices, while $\mathrm{SL}(\mathbb{R}, 2)$ be a multiplicative group of matrices with determinant 1. Is $\mathrm{SL}(\mathbb{R}, 2) \leq \mathrm{GL}(\mathbb{R}, 2)$?

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Answer. Yes, take $\mathbb{H} = \{e\} \leq \mathbb{G}$.

Question #2*

Let $\mathrm{GL}(\mathbb{R}, 2)$ be a mutliplicative group of invertable matrices, while $\mathrm{SL}(\mathbb{R}, 2)$ be a multiplicative group of matrices with determinant 1. Is $\mathrm{SL}(\mathbb{R}, 2) \leq \mathrm{GL}(\mathbb{R}, 2)$?

Answer. Yes. For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{SL}(\mathbb{R}, 2)$ the inverse is $A^{-1} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. Also, $\det(AB) = \det A \cdot \det B$, so the product of two matrices with determinant 1 has determinant 1, so the operation in closed.

Cyclic Subgroup.

Definition

Given a group $\mathbb{G}$ and $g \in \mathbb{G}$ the cyclic subgroup generated by g is

$$\langle g \rangle = \{g^n : n \in \mathbb{Z}\} = \{\ldots, g^{-3}, g^{-2}, g^{-1}, e, g, g^2, g^3, \ldots\}.$$

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Example

Consider the group of integers modulo 12, denoted by $\mathbb{Z}_{12}$.
Consider $2 \in \mathbb{Z}_{12}$, the subgroup generated by 2 is then

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Definition

We say that a group $\mathbb{G}$ is **cyclic** if there exists an element $g \in \mathbb{G}$ such that $\mathbb{G}$ is generated by g, that is, $\mathbb{G} = \langle g \rangle$.

Cyclic Subgroup Examples.

Example

Take $\mathbb{Q}^{\times}$. One of the possible cyclic subgroups is $\mathbb{H} = \{2^n : n \in \mathbb{Z}\}$.

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Question #1

What is the generator of $\mathbb{H}$ in the example above?

Question #2

What is the generator of
$7\mathbb{Z} = \{7k : k \in \mathbb{Z}\} = \{\ldots, -14, -7, 0, 7, 14, \ldots\}$?

Homomorphism

Definition

A **homomorphism** is a function $\phi : \mathbb{G} \to \mathbb{H}$ between two groups $(\mathbb{G}, \oplus)$ and $(\mathbb{H}, \odot)$ that preserves the group structure, i.e.,

$$\forall a, b \in \mathbb{G} : \phi(a \oplus b) = \phi(a) \odot \phi(b)$$

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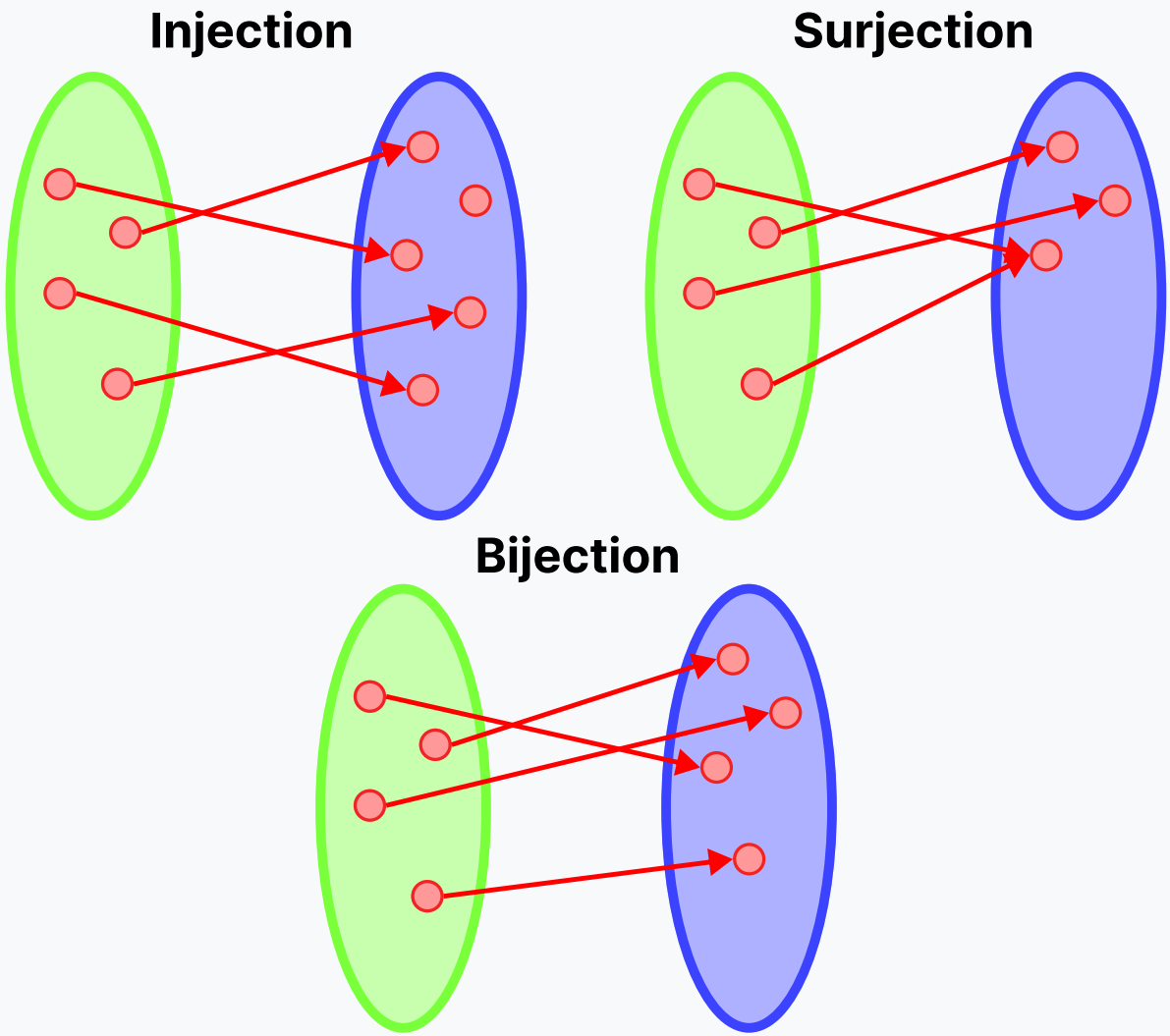
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Proof. Take any $n, m \in \mathbb{Z}$ and consider $\phi(n + m)$:

$$\phi(n + m) = 2^{n+m} = 2^n \times 2^m = \phi(n) \times \phi(m)$$

Mapping types



Homomorphism

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$\phi : k \mapsto 2^k$ from the previous example is a homomorphism between $(\mathbb{Z}, +)$ and $(\mathbb{R}_{>0}, \times)$, but not an isomorphism. Indeed, there is no $x \in \mathbb{Z}$ such that $2^x = 3 \in \mathbb{R}_{>0}$.

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Question

What can we do to make ϕ an isomorphism?

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Informal Definition

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Definition

A **field** is a set $\mathbb{F}$ with two operations $\oplus$ and $\odot$ such that:

1. $(\mathbb{F}, \oplus)$ is an abelian group with identity $e_{\oplus}$.
2. $(\mathbb{F} \setminus \{e_{\oplus}\}, \odot)$ is an abelian group.
3. The **distributive law** holds:
   $\forall a, b, c \in \mathbb{F} : a \odot (b \oplus c) = (a \odot b) \oplus (a \odot c)$.

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Example

The set $\mathbb{F}_5 = \{0, 1, 2, 3, 4\}$ with operations modulo 5 is a field.
Operation examples:

- $3 + 4 = 2$.
- $3 \times 2 = 1$.
- $4^{-1} = 4$ since $4 \times 4 = 1$.

Thank you for your attention



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